Deep Learning Framework for Camouflage Recognition in Battlefield Environments

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Proposed Architecture

- Multi-dataset based camouflage detection framework.
- Input images are collected from multiple military and camouflage datasets.
- All datasets are converted and merged into a unified YOLO format.
- YOLOv11 is used for real-time camouflage object detection.
- Multi-Modal feature extraction enhances discrimination of camouflage patterns.
- Final output includes bounding boxes and camouflage masks.

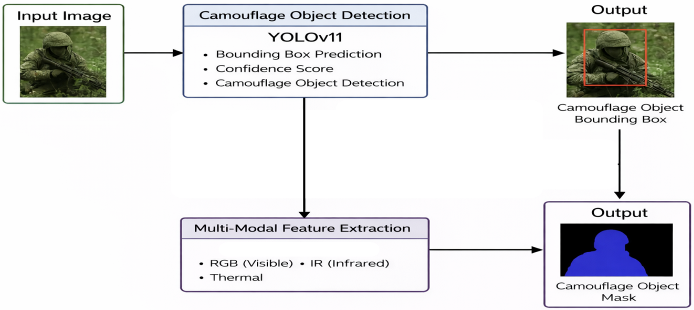

Figure 1: Architecture

System Workflow

1. Dataset collection from Kaggle repositories.
2. Conversion of all datasets into YOLO-compatible format.
3. Dataset merging and annotation validation.
4. Training of YOLOv11 model for camouflage object detection.
5. Multi-Modal feature extraction using RGB, IR, and thermal cues.
6. Generation of final detection and segmentation output.

Datasets

Table 1: Datasets Used for Object Detection

Dataset	Description
Military Assets Detection	A defense-focused dataset used for detecting military assets such as tanks, vehicles, weapons, and related equipment. It provides annotated images suitable for training YOLO-based object detection models.
VisDrone	A large-scale drone-captured dataset designed for object detection in aerial imagery. It contains complex real-world scenes with objects such as vehicles and pedestrians, useful for UAV surveillance and tracking tasks.
HIT-UAV	An aerial object detection dataset collected using UAVs, containing annotated targets such as vehicles and other objects from top-view perspectives. It is useful for training models in drone-based monitoring and defense operations.

Expected Outcome

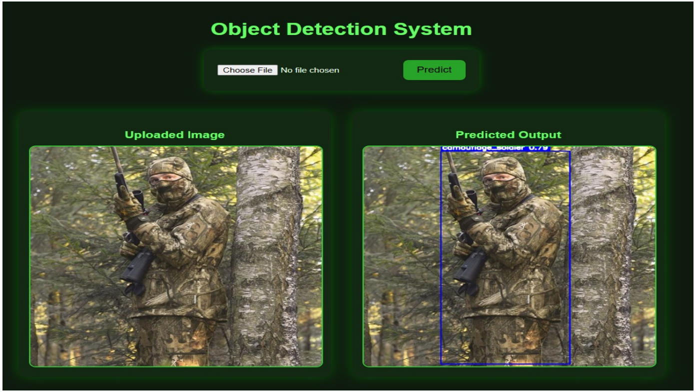

Figure 1: Detected Soldier

Thank You